

Foundation Model-Guided Discovery of Genetic Perturbations Predicted to Reverse Single-Cell Transcriptomic Aging Across Human Cell Types

Running title: Single-cell aging reversal by genetic perturbation prediction

Article type: Review and computational framework / hypothesis-generating study design

Draft status. This manuscript is formatted as a review/protocol-style scientific article. It does not report newly generated experimental or computational perturbation results. Candidate rankings are based on currently available literature and should be validated by dedicated in silico and wet-lab experiments before being presented as original results.

Keywords: single-cell RNA-seq; biological age; aging clock; transcriptomic aging; genetic perturbation; Geneformer; GEARS; scGPT; scFoundation; EZH2; E2F3; STAT3; ZFX; rejuvenation

Abstract

Biological aging is accompanied by progressive molecular changes that vary across tissues and cell types. Bulk transcriptomic or epigenetic aging clocks have provided important estimates of biological age, but they are limited by cell-composition effects and cannot directly resolve cell-type-specific aging programs. Recent advances in single-cell transcriptomics, biological age clocks, perturbation-response modeling, and foundation models create a new opportunity to identify genetic perturbations that may reverse cellular aging in a cell-type-aware manner. Here, we propose a computational framework for discovering candidate rejuvenating genetic perturbations using single-cell transcriptomic aging clocks, foundation-model-derived gene and protein representations, and perturbation-response prediction models. The framework prioritizes perturbations that reduce predicted transcriptomic biological age, shift aged cells toward young reference states, preserve cell identity, and avoid oncogenic, dedifferentiation, or pluripotency-associated signatures.

We review relevant resources, including AgeAnno, a human single-cell aging knowledgebase containing 1,678,610 cells from 28 healthy tissues across ages 0-110 years, and scAgeClock, a gated multi-head-attention single-cell transcriptomic aging clock designed to predict biological age at cellular resolution. Based on current experimental evidence, the most strongly supported candidate rejuvenation perturbations are EZH2 overexpression, E2F3 overexpression, STAT3 repression, and ZFX repression, particularly in aged human fibroblasts. These perturbations were identified in a systematic transcription-factor perturbation platform and were reported to reverse multiple aging hallmarks, including reduced proliferation, impaired proteostasis, mitochondrial dysfunction, and senescence. EZH2 overexpression was further reported to rejuvenate aged mouse liver in vivo by reversing aging-associated gene expression, reducing steatosis and fibrosis, and improving glucose tolerance.

We conclude that foundation models should be used as hypothesis-generating tools rather than definitive evidence of causal rejuvenation, because current benchmark studies indicate that deep-learning perturbation predictors do not yet consistently outperform simple linear baselines. A robust aging-reversal screen should therefore combine in silico prioritization with CRISPRa/CRISPRi Perturb-seq, single-cell biological age prediction, and functional validation.

1. Introduction

Aging is a complex biological process characterized by progressive alterations in gene regulation, chromatin state, mitochondrial function, proteostasis, inflammatory signaling, senescence, and tissue repair capacity. Although chronological age is a useful demographic variable, it does not fully capture the biological state of cells or tissues. Biological aging clocks attempt to estimate molecular age from features such as DNA methylation, gene expression, chromatin accessibility, proteomics, or multi-omic profiles. These clocks have become important tools for studying age-related disease, evaluating interventions, and identifying molecular programs associated with aging.

However, many established aging clocks are based on bulk tissue measurements. Bulk profiles are informative but confounded by changes in cell-type composition. For example, an apparently older tissue-level transcriptome may reflect an increased proportion of inflammatory or senescent cells rather than intrinsic aging of every cell type. Conversely, a perturbation may appear to rejuvenate a bulk sample by altering cell composition rather than reversing aging within a specific cell type. This limitation is particularly important for aging research, because aging affects different cell types through distinct and sometimes opposing mechanisms.

Single-cell RNA sequencing provides a more suitable framework for resolving cell-type-specific aging. By quantifying gene expression in individual cells, single-cell transcriptomics allows aging trajectories to be estimated separately for fibroblasts, hepatocytes, endothelial cells, immune cells, epithelial cells, stromal cells, and other cell populations. This resolution is essential for identifying

genetic perturbations that may reverse biological age within a defined cellular context.

Recent single-cell aging resources have made this approach increasingly feasible. AgeAnno provides a knowledgebase of aging-related single-cell annotations across human tissues, including scRNA-seq and scATAC-seq information. Its current version contains 1,678,610 cells from 28 healthy tissue samples across ages 0-110 years and includes aging-related genes, cell-type-specific analyses, regulatory information, and chromatin-accessibility features. scAgeClock further advances this field by introducing a gated multi-head-attention neural-network model for predicting human biological age at single-cell resolution from transcriptomic profiles.

In parallel, large-scale foundation models for single-cell biology have emerged. Geneformer is an attention-based model pretrained on approximately 30 million single-cell transcriptomes to enable transfer learning and context-specific network-biology predictions. scGPT is a generative pretrained transformer trained across more than 33 million cells for single-cell biology tasks. scFoundation is a large-scale foundation model covering about 20,000 genes. CellFM extends this trend by training an 800-million-parameter model on 100 million human cells. Perturbation-response models such as GEARS are specifically designed to predict transcriptional outcomes of genetic perturbations, including perturbations involving genes without prior experimental perturbation data.

These developments motivate a central question: can single-cell foundation models and perturbation-response models be used to identify genetic perturbations that reverse biological aging at the cell-type level? This manuscript proposes a framework for answering that question and synthesizes current evidence for candidate rejuvenation perturbations. We focus on perturbations represented through foundation-model-derived gene, protein, and regulatory embeddings, including approaches conceptually related to ESM-like protein representations. Because *in silico* predictions alone are not sufficient to establish rejuvenation, we emphasize rigorous validation using biological age clocks, cell identity metrics, safety filters, and functional assays.

2. Conceptual definition of single-cell aging reversal

2.1. Biological age reversal versus reprogramming

Cellular rejuvenation should be distinguished from full dedifferentiation or induced pluripotency. A perturbation that reduces a biological age clock but erases somatic identity may not be therapeutically desirable. Classical reprogramming factors such as OSK or OSKM can reverse molecular aging signatures, but they may also activate pluripotency-associated programs and tumorigenic risks when not tightly controlled. Therefore, an ideal rejuvenation perturbation should move an aged cell toward a younger version of the same cell type rather than toward an embryonic or pluripotent state.

In single-cell transcriptomic space, an ideal perturbation should satisfy four criteria:

- 1 It reduces predicted biological age.
- 2 It increases similarity to young reference cells of the same cell type.
- 3 It preserves somatic cell identity.
- 4 It avoids activation of dedifferentiation, pluripotency, DNA-damage, or oncogenic signatures.

2.2. Quantitative definition

For an aged cell or aged cell population, the primary clock-based effect can be defined as:

$$\Delta\text{Age} = \text{Age}_{\text{perturbed}} - \text{Age}_{\text{aged/control}}$$

A candidate perturbation is considered clock-rejuvenating if:

$$\Delta\text{Age} < 0$$

However, clock reduction alone is insufficient. We propose a composite rejuvenation index:

$$RI = -\Delta\text{Age} + \lambda_1\text{YS} - \lambda_2\text{LI} - \lambda_3\text{OR} - \lambda_4\text{DD} - \lambda_5\text{STR}$$

where YS is similarity to the young reference state; LI is loss of cell identity; OR is oncogenic or uncontrolled proliferation risk; DD is dedifferentiation or pluripotency-associated drift; STR is stress or DNA-damage response activation; and λ values are user-defined weights. A strong rejuvenation candidate should have a high RI, a negative ΔAge , and minimal penalties for identity loss or safety risk.

3. Data resources for single-cell aging analysis

3.1. AgeAnno

AgeAnno is one of the most relevant resources for identifying aging-related genes and regulatory programs across human cell types. It integrates single-cell RNA-seq and ATAC-seq data and provides annotations for aging-related genes across diverse tissue-cell-type contexts. In this framework, AgeAnno can be used to select age-associated genes, define young and aged reference states, identify aging-related transcription factors, examine chromatin-accessibility changes with age, and compare predicted perturbation targets against known aging-associated gene programs.

3.2. scAgeClock

scAgeClock is a single-cell transcriptome-based human aging clock model using a gated multi-head-attention neural-network architecture. It was introduced to address the lack of human aging clocks that operate at single-cell resolution while accounting for tissue and cellular heterogeneity. In the proposed framework, scAgeClock serves as the principal biological age estimator. It can be applied before and after simulated or experimental perturbation to calculate ΔAge . A candidate perturbation is prioritized if it reduces scAgeClock-predicted biological age in aged cells without causing loss of cell identity.

3.3. Human cell aging atlases

Large-scale cell atlases can provide broad reference states for cell-type-specific aging analysis. Such resources support cross-tissue validation of age-related gene programs, aging trajectories, and cell-type-specific differences in biological age. These atlases are particularly useful for determining whether a candidate perturbation is likely to be cell-type-specific or shared across several tissue contexts.

3.4. Perturb-seq datasets

Perturb-seq datasets are essential for moving from correlation to perturbation response. In a Perturb-seq experiment, cells are genetically perturbed, typically by CRISPR-based methods, and then profiled by single-cell RNA-seq. For aging-reversal discovery, the most useful Perturb-seq datasets include aged and young reference cells, genetic perturbations in aged cells, single-cell transcriptomic readouts after perturbation, sufficient biological replication, and matched functional assays.

4. Foundation models and perturbation-response models

4.1. Geneformer

Geneformer is a context-aware, attention-based deep-learning model pretrained on large-scale single-cell transcriptomes. In aging-reversal screens, Geneformer can be used to obtain cell and gene representations, estimate the effect of deleting, repressing, activating, or overexpressing genes in silico, identify genes whose perturbation shifts aged-cell embeddings toward young-cell embeddings, and prioritize candidate network regulators. However, Geneformer outputs should be interpreted as representation-level predictions rather than direct proof of biological rejuvenation.

4.2. scGPT

scGPT is a generative pretrained transformer for single-cell biology. It supports downstream tasks such as batch correction, cell-type annotation, multi-omic integration, gene network inference, and genetic perturbation prediction. For aging applications, scGPT can be used to learn cell-state representations and to model the transcriptional context in which perturbations may act. In principle, perturbations that move aged-cell embeddings closer to young-cell embeddings could be prioritized for validation.

4.3. scFoundation and CellFM

scFoundation and CellFM represent a broader trend toward large-scale pretrained models for single-cell biology. These models may be useful for extracting cell-state embeddings, gene representations, or transfer-learning features for biological age and perturbation prediction. Their primary role in this framework is to improve generalization across cell types and datasets, especially where perturbation data are sparse.

4.4. GEARS

GEARS is a geometric deep-learning model developed to predict transcriptional outcomes of genetic perturbations, including combinations of perturbations. It incorporates gene-gene relationship information through coexpression and Gene Ontology-derived knowledge graphs and can predict outcomes for perturbation sets involving genes without prior experimental perturbation data. Because aging reversal may require targeting regulatory genes that have not been tested in aging contexts, GEARS is particularly relevant for generating candidate perturbation-response profiles.

4.5. Protein-representation-informed models

The requirement that perturbations be represented through foundation-model embeddings, such as ESM-like protein representations, is biologically meaningful because gene products can be represented not only by expression values but also by protein sequence and structure-informed embeddings. Models such as SATURN demonstrate how protein language model representations can be combined with single-cell gene expression to learn biologically meaningful cell embeddings. For genetic perturbation discovery, a protein-representation-aware model could represent each candidate gene using gene expression embedding, protein sequence embedding, protein domain features, transcription-factor or chromatin-regulator annotations, gene regulatory network position, pathway membership, and prior perturbation-response data.

5. Proposed computational pipeline

5.1. Overview

The proposed pipeline consists of eight stages:

- 1 Construct young and aged single-cell reference atlases.
- 2 Estimate baseline biological age using a single-cell transcriptomic clock.
- 3 Define cell-type-specific aging vectors.
- 4 Represent candidate perturbations using foundation-model embeddings.
- 5 Predict post-perturbation transcriptomes or embeddings.
- 6 Recompute biological age after perturbation.
- 7 Apply identity and safety filters.
- 8 Prioritize candidates for experimental validation.

5.2. Construct young and aged reference states

For each cell type, cells should be divided into young and aged groups using donor chronological age. The analysis should be performed within cell types rather than across the whole tissue to avoid confounding from age-associated changes in cell-type composition. Recommended cell types for the initial screen include dermal fibroblasts, hepatocytes or hepatocyte-like cells, keratinocytes, endothelial cells, stromal cells, macrophages and monocytes, T cells, skeletal muscle satellite cells or myogenic cells, and neural or glial cell types if high-quality age-labeled datasets are available.

5.3. Estimate single-cell biological age

The biological age of each cell should be predicted using scAgeClock or a comparable single-cell transcriptomic age predictor. The baseline age gap for each cell type can be summarized as:

$$\text{AgeGap}_{\text{celltype}} = \text{median}(\text{Age}_{\text{aged}}) - \text{median}(\text{Age}_{\text{young}})$$

Only cell types with a robust and reproducible age gap should be included in perturbation prioritization.

5.4. Define aging vectors

Aging can be represented as a vector in transcriptomic or embedding space:

$$\mathbf{V}_{\text{aging}} = \mu_{\text{aged}} - \mu_{\text{young}}$$

where μ_{aged} and μ_{young} are mean representations of aged and young cells of the same cell type. A candidate perturbation is desirable if its predicted effect vector points opposite to the aging vector:

$$\text{cosine}(\mathbf{V}_{\text{perturbation}}, -\mathbf{V}_{\text{aging}}) > 0$$

5.5. Generate candidate perturbation representations

Each candidate gene should be represented using multiple feature types:

$$\mathbf{G}_i = [\mathbf{E}_{\text{gene}}, \mathbf{E}_{\text{protein}}, \mathbf{E}_{\text{regulatory}}, \mathbf{E}_{\text{network}}, \mathbf{E}_{\text{prior}}]$$

Candidate perturbations should include both activation and repression:

$$\mathbf{P}_i \in \{\text{Gene}_i \uparrow, \text{Gene}_i \downarrow\}$$

For transcription factors and chromatin regulators, both directions should be considered because overexpression and repression can have opposite effects.

5.6. Predict post-perturbation states

Post-perturbation states can be predicted using one or more models, including Geneformer for embedding-level perturbation shifts, GEARS for predicted transcriptional response, scGPT or scFoundation for cell-state and gene-level representation shifts, and custom perturbation-response models trained on Perturb-seq data. The output should ideally be a predicted transcriptome that can be passed into a transcriptomic aging clock.

5.7. Score biological age reversal and safety

For each candidate perturbation, biological age reduction is calculated as:

$$\Delta\text{Age}_i = \text{Age}(\mathbf{X}_{\text{perturbed},i}) - \text{Age}(\mathbf{X}_{\text{aged}})$$

Perturbations are ranked by negative ΔAge only after applying filters for identity and safety. A candidate is excluded or penalized if it reduces expression of canonical cell identity markers, increases pluripotency-associated programs, strongly activates cell-cycle or oncogenic signatures without functional control, increases DNA damage response, induces stress-response artifacts, or

shifts the cell toward an unrelated lineage.

6. Evidence synthesis: candidate rejuvenation perturbations

6.1. Highest-priority perturbations

The strongest currently available evidence supports four single transcription-factor perturbations in aged human fibroblasts: EZH2 overexpression, E2F3 overexpression, STAT3 repression, and ZFX repression. A systematic transcription-factor perturbation platform reported that overexpression of E2F3 or EZH2 and repression of STAT3 or ZFX reversed cellular hallmarks of aging, including proliferation decline, proteostasis impairment, mitochondrial dysfunction, and senescence.

Among these, EZH2 overexpression is the highest-priority candidate because it was also reported to rejuvenate aged mouse liver in vivo, reversing aging-associated gene expression and improving liver phenotypes including steatosis, fibrosis, and glucose tolerance.

6.2. Secondary candidates

SRSF1 overexpression is a promising secondary candidate. A transcriptomic reprogramming screen reported that SRSF1 induction shifted cellular transcriptomes toward a younger state and improved several aging-associated phenotypes, including senescence, proteasome function, collagen production, oxidative stress response, and wound-healing dysfunction.

SB000 is another potentially important candidate, but it should be interpreted cautiously. A 2025 preprint reported that SB000 drove multi-omic rejuvenation in human fibroblasts, including reversal of epigenetic clocks, reduced single-cell transcriptomic age, and decreased senescence-associated gene expression. However, because the target identity is not sufficiently transparent for independent implementation, SB000 should not yet be treated as an actionable open target.

7. Prioritized candidate list

Table 1. Candidate genetic perturbations for single-cell transcriptomic aging reversal.

Rank	Candidate perturbation	Primary supported context	Evidence level	Rationale	Key concerns
1	EZH2 overexpression / CRISPRa	Aged human fibroblasts; aged mouse liver	High	Reverses fibroblast aging hallmarks and has in vivo aged-liver evidence.	Potential cancer/proliferation risk; context-dependent chromatin effects
2	E2F3 overexpression / CRISPRa	Aged human fibroblasts	High for fibroblasts	Reverses aging-associated functional decline in fibroblast model.	Cell-cycle activation; oncogenic-risk monitoring required
3	STAT3 repression / CRISPRi	Aged human fibroblasts	High for fibroblasts	Reverses aging hallmarks in fibroblast model.	STAT3 has major immune, inflammatory, and regenerative roles
4	ZFX repression / CRISPRi	Aged human fibroblasts	Moderate-high	Validated in fibroblast rejuvenation screen.	Possible effects on stemness and cell identity
5	SRSF1 overexpression	Fibroblast and wound-healing models	Moderate; preprint-level	Reported to shift transcriptome toward young state and improve aging phenotypes.	Requires peer-reviewed and cell-type-specific validation
6	SB000 overexpression	Human fibroblasts; claimed multi-cell-type effects	Promising but not fully actionable	Reported multi-omic age reversal and identity preservation.	Target identity not sufficiently transparent; independent validation needed

8. Cell-type-specific prioritization

Table 2. Suggested first-pass perturbation priorities by cell type.

Cell type	First-line candidates	Secondary / exploratory candidates	Key readouts
Fibroblasts	EZH2↑, E2F3↑, STAT3↓, ZFX↓	SRSF1↑; SB000 when target is disclosed	Clock age, fibroblast identity, CDKN1A/CDKN2A, ECM/collagen, proteostasis, mitochondria, proliferation, DNA damage
Hepatocytes / liver	EZH2↑	E2F3↑ and STAT3↓ as exploratory comparators	Hepatocyte age, lipid metabolism, fibrosis genes, inflammation, hepatocyte identity, tumor-risk signatures
Keratinocytes / skin epithelial cells	EZH2↑, SRSF1↑	SB000 if target becomes independently testable	Keratinocyte identity, barrier programs, senescence, wound-healing and inflammatory programs
Endothelial cells	No validated universal target	EZH2↑ or STAT3↓ as hypotheses only	Endothelial identity, angiogenesis, inflammatory activation, senescence, nitric-oxide/vascular homeostasis genes
Immune cells	No universal target currently supported	Cell-type-specific in silico screen required	Activation/exhaustion state, inflammatory tone, naive-memory balance, cell identity, clock age
Neural and glial cells	No direct recommendation	Use cautious cell-type-specific screening	Avoid cell-cycle re-entry, monitor neuronal/glial identity, stress and DNA-damage signatures

9. Mechanistic interpretation of top candidates

9.1. EZH2

EZH2 encodes the catalytic component of Polycomb Repressive Complex 2 and is involved in H3K27me3-associated transcriptional repression. Its emergence as a top rejuvenation candidate suggests that restoration of youthful chromatin regulation may reverse part of the aged cellular state. The reported in vivo liver rejuvenation effect makes EZH2 particularly compelling. However, EZH2 also has context-dependent roles in cancer biology, proliferation, and cell fate. Therefore, EZH2 activation should be evaluated under strict safety filters. A favorable EZH2 response would reduce biological age without inducing uncontrolled proliferation, DNA damage, or loss of cell identity.

9.2. E2F3

E2F3 is a transcription factor involved in cell-cycle regulation. Its overexpression may reverse aging phenotypes partly by restoring proliferative capacity and metabolic function in aged fibroblasts. However, this same property raises safety concerns. E2F3 should therefore be tested with controlled expression levels, short induction windows, and careful oncogenic-risk assessment.

9.3. STAT3

STAT3 is a transcription factor involved in cytokine signaling, inflammation, stress response, regeneration, and immune regulation. STAT3 repression was reported to reverse aging hallmarks in fibroblasts. Mechanistically, this may reflect suppression of chronic stress or inflammatory transcriptional programs. However, because STAT3 can also be protective or regenerative in some tissues, its effects should be evaluated in a cell-type-specific manner.

9.4. ZFX

ZFX is a transcriptional regulator with context-dependent roles in cell state and stemness. Its repression was validated as a rejuvenation-associated perturbation in aged fibroblasts. Because ZFX may influence cell identity or stem-like programs, any ZFX-targeting strategy should include strong identity-preservation and dedifferentiation filters.

9.5. SRSF1

SRSF1 is a splicing factor and transcriptomic regulator. Its reported ability to shift aged cells toward a younger transcriptomic state suggests that age-related RNA-processing changes may be therapeutically targetable. SRSF1 is particularly interesting for skin, fibroblast, and wound-healing contexts, but it currently has weaker evidence than EZH2, E2F3, STAT3, and ZFX.

10. Safety and interpretability considerations

10.1. Biological clock reversal is not sufficient

Aging clocks are predictive models, not complete definitions of biological function. A perturbation may reduce predicted age by altering a subset of clock genes without improving cell function. Therefore, biological age reduction must be paired with functional readouts.

10.2. Preservation of cell identity

A rejuvenated fibroblast should remain a fibroblast. A rejuvenated hepatocyte should remain a hepatocyte. Loss of identity may produce an artificially young transcriptomic profile but should not be considered successful rejuvenation.

10.3. Avoidance of pluripotency and dedifferentiation

Partial reprogramming can reverse molecular age but may activate developmental or pluripotency-associated programs. A safe rejuvenation candidate should avoid strong induction of pluripotency markers and should not shift cells toward embryonic-like states.

10.4. Oncogenic risk

Several top candidates, especially EZH2 and E2F3, are linked to proliferation and cancer biology in some contexts. Their use as rejuvenation factors therefore requires careful dose control, temporal control, and long-term safety assessment.

10.5. Foundation-model limitations

Although foundation models are powerful representation-learning systems, they do not necessarily encode causal regulatory logic. Benchmarking work has shown that deep-learning-based gene perturbation effect prediction models do not consistently outperform simple linear baselines. This finding is critical: foundation models should guide candidate selection, but experimental validation remains essential.

11. Proposed validation experiments

11.1. Phase I: aged human dermal fibroblast validation

The first experimental phase should use aged human dermal fibroblasts because this is the best-supported cell type. Perturbations should include CRISPRa-EZH2, CRISPRa-E2F3, CRISPRi-STAT3, CRISPRi-ZFX, SRSF1 overexpression, non-targeting control, and a positive reprogramming control if appropriate and safely controlled.

The primary endpoint should be reduction in single-cell transcriptomic biological age. Secondary endpoints should include young-state transcriptomic similarity, fibroblast identity preservation, reduction in senescence-associated markers, improved proteostasis, improved mitochondrial function, controlled proliferation, absence of pluripotency activation, and absence of excessive oncogenic or DNA-damage signatures.

11.2. Phase II: liver-focused validation

Because EZH2 has in vivo aged-liver evidence, the second phase should test EZH2 activation in hepatocyte-like systems or primary hepatocytes. Perturbations should include EZH2 overexpression or CRISPRa-EZH2, E2F3 activation as an exploratory comparator, STAT3 repression as a context-dependent comparator, and non-targeting control. The primary endpoint should be reduction in hepatocyte transcriptomic biological age. Secondary endpoints should include restoration of lipid metabolism, reduced inflammatory and fibrosis-associated programs, preserved hepatocyte identity, and absence of uncontrolled proliferation.

11.3. Phase III: cell-type expansion

After fibroblast and liver validation, the framework can be expanded to keratinocytes, endothelial cells, macrophages, T cells, skeletal muscle cells, and neural or glial cell types. For each cell type, the perturbation ranking should be recomputed rather than transferred directly from fibroblasts.

12. Proposed figures

Figure 1. Overview of the computational framework. A schematic showing young and aged single-cell transcriptomic atlases as input. Each cell is annotated by cell type and assigned a biological age using a single-cell transcriptomic aging clock. Candidate genes are represented using foundation-model-derived gene embeddings, protein sequence embeddings, regulatory features, and gene-network features. Perturbation-response models predict post-perturbation cell states. Candidate perturbations are ranked by biological age reduction, young-state similarity, cell identity preservation, and safety filters.

Figure 2. Definition of rejuvenation in single-cell transcriptomic space. A conceptual UMAP or embedding diagram showing young fibroblasts, aged fibroblasts, and perturbed aged fibroblasts. A successful rejuvenation perturbation moves aged fibroblasts toward the young fibroblast state while retaining fibroblast identity. A failed perturbation either does not move the cells, moves them toward an unrelated lineage, or induces dedifferentiation or pluripotency.

Figure 3. Candidate gene prioritization. A ranked heatmap showing predicted or evidence-supported effects of EZH2↑, E2F3↑, STAT3↓, ZFX↓, SRSF1↑, and SB000 across fibroblasts, hepatocytes, keratinocytes, endothelial cells, and immune cells. Evidence strength is highest for fibroblasts and EZH2 liver validation, while other cell-type predictions are marked as exploratory.

Figure 4. Safety filtering strategy. A multi-axis scoring plot showing biological age reduction, young-state similarity, cell identity preservation, oncogenic-risk score, pluripotency score, and stress-response score. Ideal candidates occupy the region of strong age reduction, high identity preservation, and low safety risk.

Figure 5. Experimental validation workflow. A workflow showing CRISPRa/CRISPRi perturbation in aged cells, single-cell RNA-seq readout, biological age prediction, functional assays, and final target prioritization.

13. Proposed supplementary tables

Supplementary Table 1. Single-cell aging datasets

Recommended columns: dataset name; tissue; cell types; age range; number of cells; modality; availability; suitability for clock training; suitability for perturbation validation.

Supplementary Table 2. Candidate perturbation genes

Recommended columns: gene; perturbation direction; known biological role; supported cell type; evidence type; expected rejuvenation mechanism; safety concerns; validation priority.

Supplementary Table 3. Aging and safety gene signatures

Recommended categories: senescence; DNA damage; mitochondrial dysfunction; inflammatory signaling; proteostasis; extracellular matrix; cell identity markers; pluripotency markers; oncogenic signatures.

14. Discussion

This manuscript proposes a framework for identifying genetic perturbations that may reverse cellular aging using single-cell transcriptomic clocks and foundation-model-guided perturbation prediction. The most important conceptual point is that rejuvenation should be defined not merely as a reduction in predicted age, but as a coordinated movement of aged cells toward a younger version of the same cell type. This distinction prevents misclassification of dedifferentiation, lineage drift, or uncontrolled proliferation as successful rejuvenation.

The strongest current evidence supports $EZH2\uparrow$, $E2F3\uparrow$, $STAT3\downarrow$, and $ZFX\downarrow$ in aged human fibroblasts. Among these, $EZH2\uparrow$ is the most compelling because it has both in vitro fibroblast evidence and in vivo aged-liver evidence. $E2F3\uparrow$ is also promising but requires careful monitoring because of its role in cell-cycle control. $STAT3\downarrow$ may reduce inflammatory or stress-associated aging programs, but its effects are likely to be highly context-dependent. $ZFX\downarrow$ is validated in fibroblasts but requires careful cell-identity and stemness monitoring.

$SRSF1\uparrow$ is a promising secondary candidate, especially for fibroblast and skin-associated aging, because transcriptomic reprogramming studies suggest that it may shift aged cells toward younger states and improve wound-healing-related phenotypes. SB000 is potentially important but should remain in a separate category until the target identity is fully disclosed and independently validated.

Foundation models can accelerate this discovery process by ranking candidate perturbations, especially when combined with protein representations, regulatory-network information, and perturbation-response models. However, current evidence does not justify treating foundation-model predictions as definitive. Benchmarking work has shown that deep-learning perturbation predictors do not yet consistently outperform simple linear baselines. Thus, the role of foundation models should be to prioritize hypotheses, not to replace perturbation experiments.

A practical path forward is to begin with aged human fibroblasts, validate the top four perturbations using CRISPRa/CRISPRi Perturb-seq, score biological age reversal with scAgeClock or related clocks, and then expand to hepatocytes, keratinocytes, endothelial cells, and immune cells. Each expansion should recompute cell-type-specific aging vectors and perturbation rankings rather than assuming universal rejuvenation targets.

15. Limitations

This article is a synthesis and computational framework rather than an original experimental report. The proposed ranking is based on published and preprint evidence, not on newly generated perturbation data. Therefore, the candidate list should be interpreted as a prioritized hypothesis set.

A second limitation is that single-cell transcriptomic age is only one dimension of biological age. Transcriptomic rejuvenation should be compared with DNA methylation age, chromatin accessibility, proteomic changes, mitochondrial function, senescence burden, and tissue-level function.

A third limitation is that model-predicted perturbation effects may not generalize across datasets, donors, species, or experimental platforms. Batch effects, cell culture conditions, donor age, sex, tissue source, and disease status may all influence perturbation response.

Finally, candidate perturbations such as $EZH2\uparrow$ and $E2F3\uparrow$ raise safety concerns because they may influence proliferation or cancer-associated programs. These perturbations should not be interpreted as therapeutic recommendations without extensive validation.

16. Conclusion

Single-cell transcriptomic aging clocks and foundation models create a powerful framework for identifying genetic perturbations that may reverse biological aging in a cell-type-specific manner. The best-supported current candidates are:

$$\text{EZH2}\uparrow > \text{E2F3}\uparrow > \text{STAT3}\downarrow > \text{ZFX}\downarrow > \text{SRSF1}\uparrow$$

The strongest evidence supports these perturbations in aged human fibroblasts, with EZH2 $\uparrow$ additionally supported by aged-liver in vivo evidence. SB000 is a promising but currently non-transparent candidate that requires independent validation.

The central recommendation is that future aging-reversal discovery should combine single-cell biological age clocks, young-versus-aged cell-type-specific reference atlases, foundation-model-derived gene and protein representations, perturbation-response prediction, strict identity and safety filters, CRISPRa/CRISPRi Perturb-seq validation, and functional assays. This integrated strategy can move the field from descriptive aging signatures toward causal, cell-type-specific rejuvenation interventions.

Data availability

No new experimental dataset was generated in this manuscript. The proposed framework relies on publicly available single-cell aging resources, including AgeAnno, scAgeClock-associated resources, public single-cell atlases, and published or preprint perturbation studies.

Code availability

No new code is provided in this manuscript draft. A reproducible implementation should include scripts for single-cell preprocessing, biological age prediction, foundation-model embedding extraction, perturbation-response prediction, rejuvenation scoring, and statistical validation.

Author contributions

Conceptualization: [Author names]. Literature synthesis: [Author names]. Computational framework design: [Author names]. Manuscript writing: [Author names]. Critical revision: [Author names].

Competing interests

The authors declare no competing interests. If any author is affiliated with a biotechnology company or has intellectual property related to aging clocks, genetic perturbation, or cellular rejuvenation, this section should be updated accordingly.

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